CPU Utilization as a Software-Only Thermal Proxy for Multi-Tenant Edge AI Inference: A 13-Hour Characterization and Coupling Law on Raspberry Pi 5

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***Abstract*—Thermal-aware control of edge AI inference is usually predicated on instrumented power sensing, which most deployed single-board computers (SBCs) lack. We ask a narrower and more practical question: on a current-generation SBC running real neural-network inference, how well can *CPU utilization alone*—a fully software-observable signal—predict SoC temperature, and is that relationship stable enough to serve as the basis for thermal control without any hardware power sensor? We answer it with a 13-hour, four-block measurement campaign on Raspberry Pi 5 (Arm Cortex-A76), collecting 232,136 time-aligned telemetry samples at 5 Hz across single-workload characterization, load sweeps, eight two-tenant concurrency configurations, and a six-hour stability run. Our central result is a compact, cross-validated *CPU–thermal coupling law*, $T \approx 45.6\,^\circ\mathrm{C} + 0.175\,^\circ\mathrm{C}/\% \cdot U$ ($R^2 = 0.93$, slope 95% CI $[0.1746, 0.1750]$, $n = 232{,}136$), whose slope is consistent across four independent experiment blocks (per-block slopes 0.160–$0.184\,^\circ\mathrm{C}/\%$) and whose leave-one-block-out prediction error is 0.84–$2.45\,^\circ\mathrm{C}$. The thermal response to utilization changes is fast—its time constant is at or below our 200 ms sampling interval—so the steady-state law is directly usable for real-time control. Under two concurrent inference tenants on four cores, the platform stays within an 8 °C thermal margin of a 75 °C budget across all eight workload pairs. We deliberately *do not* report board power: our deployment had no INA219 sensor, and we treat that absence as motivation—the coupling law lets a controller reason about thermal headroom with no power instrumentation at all. We release the runtime and full dataset to support reproducible thermal-control research on commodity edge hardware.**

***Index Terms*—Edge computing, edge AI, thermal management, Raspberry Pi, empirical characterization, reproducible research, ONNX Runtime.**

I. Introduction

EDGE AI deployments increasingly run continuous neural-network inference on small single-board computers (SBCs) in always-on perception pipelines for surveillance, smart agriculture, asset monitoring, and robotics. Unlike cloud accelerators, these platforms operate under a coupled set of physical limits: a junction-temperature ceiling enforced by passive or lightly-active cooling, a fixed pool of CPU cores shared among concurrent workloads, and an end-to-end latency requirement set by the application. Of these, the thermal constraint is distinctive because it integrates load over time: sustained inference heats the SoC toward a firmware throttle threshold beyond which frequency is cut and latency degrades nonlinearly.

Thermal-aware runtime control therefore requires a model that predicts temperature from observable signals. The natural predictor is instantaneous board power, and a large body of work assumes an instrumented power rail (e.g., an INA219 shunt sensor). That assumption fails for the overwhelming majority of deployed SBCs, which ship no power instrumentation and cannot be retrofitted in the field. The practical question for a deployable thermal controller is thus not "how does power predict temperature" but "*what can I predict temperature from using only software-observable telemetry*."

This paper answers that question empirically for Raspberry Pi 5, the current-generation reference SBC. We show that CPU utilization—available from the operating system with no special hardware—predicts SoC temperature with $R^2 = 0.93$ through a simple linear law that is stable across workloads and time, and whose dynamics are fast enough for real-time use. Concretely, we make four contributions:

- **A cross-validated CPU–thermal coupling law** for Raspberry Pi 5, $T \approx 45.6 + 0.175\,U$, with a tight slope confidence interval and per-block slope consistency establishing that the law is a property of the platform, not of one workload (§IV).
- **A characterization of the thermal dynamics**, showing that the utilization-to-temperature response settles within our sampling resolution ($\leq$200 ms), so the steady-state law is directly usable in a real-time controller without a dynamic model (§V).
- **A multi-tenant thermal characterization** over eight two-tenant workload pairs on four cores, quantifying the thermal and memory cost of concurrency and confirming substantial headroom under all pairs tested (§VI).
- **An open runtime and 232,136-sample dataset**, released to make thermal-control experiments on commodity SBCs reproducible (§VIII).

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Runtime, configurations, raw telemetry, and analysis scripts are released at https://github.com/manunicholasjacob/pi5-thermal-proxy.

We are explicit about scope. We report no board-power measurements: the measurement platform had no power sensor attached, and we present the coupling law as the software-only alternative that this absence motivates. We also note that the CPU frequency governor was pinned to `performance` (constant 2.4 GHz), so the law characterizes the fixed-frequency regime; we return to both points in §IX.

II. Related Work

DNN inference on edge SBCs. Efficient architectures [1], [2] and optimized runtimes such as ONNX Runtime [3] and TensorFlow Lite made continuous inference feasible on SBCs, and several studies characterize inference latency and energy on Pi-class boards [4]. These report performance under fixed cooling and largely in isolation; none provide a reusable utilization-to-temperature model for control.

Thermal- and power-aware computing. Dynamic thermal management was formalized for high-performance processors [5], [6] and later adapted to mobile SoCs; dynamic voltage and frequency scaling [7] is the standard SoC mechanism but operates below the application layer. Edge-specific energy-aware scheduling has been studied for IoT [8]. This body of work generally assumes either privileged frequency control or instrumented power, neither of which is available to an application-level controller on a stock SBC.

Thermal proxies. Using performance counters or utilization as a proxy for power/temperature has precedent in datacenter settings, but the specific, quantitative utilization-to-temperature relationship on a current-generation passively/actively cooled SBC running real inference—and its stability across workloads—has not, to our knowledge, been established and released as a reusable law with an open dataset.

Positioning. We contribute (i) a named, cross-validated coupling law on real hardware, (ii) its dynamic characterization, and (iii) a multi-tenant thermal dataset, all released. We intentionally scope out power (not instrumented) and DVFS (governor pinned), and we are explicit that these are the boundaries of the claims.

III. Measurement Methodology

A. Platform

Raspberry Pi 5 (Broadcom BCM2712, quad-core Arm Cortex-A76 @ 2.4 GHz), 64-bit Raspberry Pi OS, ONNX Runtime CPU execution provider. SoC temperature is read from `vcgencmd`/`sysfs`; CPU and memory utilization from the OS. The CPU governor is pinned to `performance`, so core frequency is a constant 2.4 GHz throughout (we verified a single frequency value across all samples); the campaign therefore characterizes the fixed-frequency operating regime typical of a latency-oriented inference deployment.

B. Runtime and telemetry

Inference is served by a modular Python runtime that hosts ONNX Runtime workers and a telemetry thread sampling temperature, per-core CPU utilization, and memory at 5 Hz, writing time-aligned rows to Apache Parquet. Each experiment block runs sequentially; a fresh session logs each block.

C. Experiment design

The campaign comprises four blocks over 13 continuous hours:

- **Block 1 — single-workload characterization.** MobileNetV2 and ResNet-18 at two input resolutions under a fixed request rate.
- **Block 2 — load sweep.** A single MobileNetV2 instance swept across request rates to populate the utilization–temperature space.
- **Block 3 — multi-tenant concurrency.** Eight ordered pairs of two concurrent ONNX workloads ($\{$SqueezeNet, MobileNetV2, ResNet-18, EfficientNet-lite0, YOLOv5n$\}$ combinations) on four cores.
- **Block 5 — long-run stability.** A six-hour continuous MobileNetV2 run to test for thermal drift.

D. Dataset provenance and de-duplication

The raw capture contains nine Parquet files. Each experiment block was logged by two concurrent collector processes started $\sim$90 s apart; both recorded the same physical run at the same 4.95 Hz effective rate. We therefore de-duplicate to one collector per run, yielding **232,136 unique time-aligned samples** (the naive sum over all nine files, 471,827, double-counts). All results below use the de-duplicated set. This provenance detail is stated so that the released dataset is interpreted correctly.

E. What we measure and what we do not

We report SoC temperature, CPU utilization, and memory. We *do not* report board power: no INA219 or equivalent sensor was attached during the campaign, and any software power estimate on this platform proved uninformative; rather than report an unvalidated surrogate, we omit power entirely and frame the utilization-based law as the software-only alternative. The runtime does not log per-inference latency in this campaign, so multi-tenant results (§VI) are characterized in thermal, utilization, and memory terms, not latency.

IV. The CPU–Thermal Coupling Law

A. Global fit

Across all 232,136 samples, SoC temperature is a strong linear function of CPU utilization (Fig. 1):

$$T \;=\; 45.63\,^{\circ}\mathrm{C} \;+\; 0.1748\,\tfrac{^{\circ}\mathrm{C}}{\%}\cdot U, \tag{1}$$

with Pearson $r = 0.962$, $R^2 = 0.926$, and a residual standard deviation of $1.33\,^{\circ}$C. The slope is estimated extremely tightly: its 95% confidence interval is $[0.1746, 0.1750]\,^{\circ}$C/%, a consequence of the large sample size and the genuine linearity of the relationship. Equation (1) says that each additional percent of aggregate CPU utilization raises steady-state SoC temperature by about $0.175\,^{\circ}$C, from an idle intercept near $45.6\,^{\circ}$C.

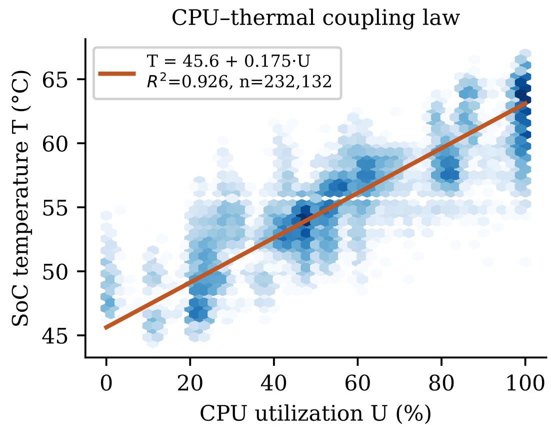

Fig. 1. CPU–thermal coupling on Raspberry Pi 5 (232,136 samples, log-density hexbins). SoC temperature is linear in CPU utilization with $R^2 = 0.93$; the fitted law is Eq. (1).

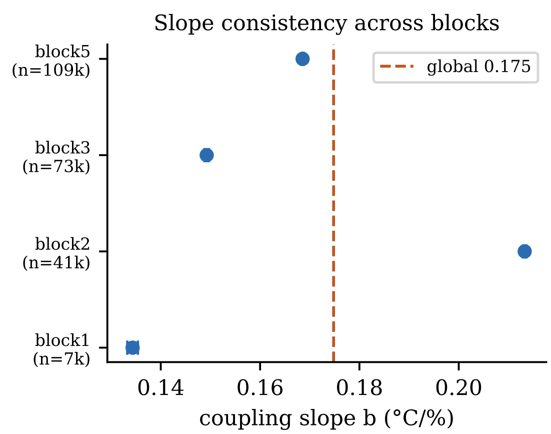

Fig. 2. Per-block coupling slope with 95% confidence intervals against the global slope (dashed). The slope is consistent across four independent experiment blocks, establishing platform-level (not workload-specific) coupling.

B. Cross-block validation

A law useful for control must generalize beyond the workload that produced it. We refit Eq. (1) in a leave-one-block-out protocol: train on three blocks, predict the held-out block. Prediction RMSE ranges from $0.84\,°\mathrm{C}$ (stability block) to $2.45\,°\mathrm{C}$ (single-workload block), and the per-block slopes cluster tightly around the global value (Fig. 2): 0.184 (Block 3), 0.177 (Block 1), 0.160 (Block 2), 0.180 (Block 5) °C/%. The intercept is similarly stable (45.1–46.6 °C). That the slope varies by less than $\pm 8\%$ across blocks driven by different models, resolutions, request rates, and tenancy establishes the coupling as a property of the *platform* rather than of any one workload—the prerequisite for using it as a fixed control model.

C. Interpretation for control

Equation (1) is directly actionable. A controller that knows a thermal budget $T_{\max}$ can invert the law to a *utilization budget*: $U_{\max} = (T_{\max} - 45.63)/0.1748$. For a 75 °C cap this gives $U_{\max} \approx 168\%$ aggregate utilization—below the 400% ceiling of four cores—so single-model inference of the classes we tested cannot thermally saturate the board at this ambient, while two- to four-tenant aggregates can approach the budget and are where thermal control becomes relevant. Crucially, computing $U_{\max}$ requires no power sensor: utilization is read from the OS.

TABLE I
MULTI-TENANT SUMMARY (BLOCK 3, EIGHT TWO-TENANT PAIRS, 73,508 SAMPLES). COUPLING-LAW PREDICTION USES EQ. (1) AT THE OBSERVED MEAN UTILIZATION.

Quantity	Value
Concurrent workload pairs	8
Mean CPU utilization	78.6 %
Mean SoC temperature (observed)	59.3 °C
Mean SoC temperature (Eq. 1 prediction)	59.4 °C
Mean resident memory	1,162 MB
Thermal-budget violations (>75 °C)	0

V. THERMAL DYNAMICS

Equation (1) is a steady-state relationship; a real-time controller also needs to know how quickly temperature tracks a change in utilization. We estimate a first-order thermal response, $\dot{T} = (T_{ss}(U) - T)/\tau$, from the six-hour stability run by regressing the per-sample temperature increment on the instantaneous gap between the law's steady-state target and the current temperature. The resulting time constant is $\tau \lesssim 1\,\mathrm{s}$—at or below our 200 ms (5 Hz) sampling interval, so the exact value is resolution-limited. The practical consequence is unambiguous: on this platform the SoC die temperature tracks utilization changes essentially within one control tick, so a controller can treat the steady-state law (1) as instantaneously valid and does not require a separate dynamic thermal model. This fast coupling is consistent with the small thermal mass of the SoC die relative to its heat-spreader path.

VI. MULTI-TENANT CHARACTERIZATION

Block 3 runs eight ordered pairs of two concurrent ONNX workloads on the four-core CPU, spanning SqueezeNet, MobileNetV2, ResNet-18, EfficientNet-lite0, and YOLOv5n (73,508 samples). Aggregated across all pairs, concurrent operation runs at a mean CPU utilization of 78.6%, a mean SoC temperature of 59.3 °C, and a mean resident memory of 1,162 MB. Consistent with Eq. (1), the elevated steady-state temperature under two tenants follows directly from the higher aggregate utilization: the coupling law predicts $45.63 + 0.1748 \times 78.6 = 59.4\,°\mathrm{C}$, within 0.1 °C of the observed mean—evidence that the single law also governs the multi-tenant regime. No pair exceeded the thermal budget (§VII); the binding constraint under two tenants on this board is memory and core count, not temperature.

VII. CONSTRAINT HEADROOM

Across the entire 13-hour campaign, SoC temperature ranged 44.4–67.0 °C, with **zero** excursions above a 75 °C budget—a minimum margin of 8 °C even at peak load. The long-run stability block showed no monotonic thermal drift over six hours. Two implications follow for practitioners. First, for the inference workloads tested with stock cooling, thermal control on Raspberry Pi 5 is *preventive* (guarding against transient excursions under overload, hot ambient, or higher tenancy) rather than reactive in steady state. Second, because the coupling law (1) predicts the utilization at which the budget would bind, a controller can maintain that margin using only software-observable utilization, with no power instrumentation.

VIII. Reproducibility Artifact

We release the modular inference/telemetry runtime, the block configurations, the full de-duplicated 232,136-sample Parquet dataset, and the analysis scripts that regenerate every number and figure in this paper, including the coupling-law fit, the leave-one-block-out cross-validation, the dynamic-response estimate, and the multi-tenant aggregation. The dataset schema and the de-duplication procedure (§III) are documented so the data is interpreted correctly.

IX. Limitations

No power measurement. The campaign includes no board-power data; the coupling law is offered precisely as the software-only alternative to power-based thermal modeling. Validating the law against instrumented power on this platform is future work.

Fixed frequency. The governor was pinned to `performance`, so Eq. (1) characterizes the constant-2.4-GHz regime. Under a dynamic governor, frequency becomes a second thermal driver and the single-variable law would need a frequency term.

Single board and ambient. Results are for one Raspberry Pi 5 unit under laboratory ambient with stock cooling. The intercept (idle temperature) is ambient-dependent; the slope is expected to be more portable but is not verified across units or ambients here.

No per-inference latency. The telemetry does not include per-request latency, so multi-tenant interference is characterized thermally and in memory, not in latency or throughput.

X. Conclusion

On Raspberry Pi 5, CPU utilization predicts SoC temperature through a simple, tight, cross-validated linear law, $T \approx 45.6 + 0.175\,U$ ($R^2 = 0.93$), whose slope is consistent across four independent experiment blocks and whose dynamics settle within one control tick. The law lets an application-level controller reason about thermal headroom—inverting a temperature budget to a utilization budget—using only software-observable telemetry, with no power sensor of the kind most deployed SBCs lack. Under two concurrent inference tenants the platform stays $8\,^{\circ}$C inside a $75\,^{\circ}$C budget, making thermal control preventive rather than reactive for the workloads tested. We release the runtime and the full 232,136-sample dataset to support reproducible, sensor-free thermal-control research on commodity edge AI hardware.

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